Effect of a 12-Week Structured Walking Program on Glycemic Control in Adults with Type 2 Diabetes: A Randomized Controlled Trial

A. R. Mehta, S. Kapoor, J. Lindgren, P. N. Rao
Department of Endocrinology, Riverside Community Health Institute

Abstract

Background. Physical activity is a cornerstone of type 2 diabetes management, yet structured, low-cost interventions remain underused in routine primary care. We evaluated whether a supervised, progressive walking program improves glycemic control compared with usual care.

Methods. In this single-center, parallel-group randomized controlled trial, 122 adults (aged 40–68 years) with HbA1c between 7.5% and 9.0% were randomized 1:1 to a 12-week walking program (target 150 min/week, progressive intensity) or usual care. The primary outcome was change in HbA1c at week 12. Secondary outcomes included body weight, BMI, systolic blood pressure, and fasting glucose. Analysis followed the intention-to-treat principle.

Results. The walking group showed a mean HbA1c reduction of 0.9 percentage points versus 0.1 in usual care (between-group difference –0.8%, 95% CI –1.0 to –0.6, p<0.001). Significant improvements were also seen in body weight (–3.4 vs –0.6 kg) and fasting glucose (–18.5 vs –3.2 mg/dL). Adherence was 84%, and no serious adverse events were attributed to the intervention.

Conclusions. A simple, structured walking program produced clinically meaningful improvements in glycemic control over 12 weeks and represents a scalable option for primary-care settings.

Keywords: type 2 diabetes, physical activity, HbA1c, walking, randomized controlled trial, lifestyle intervention

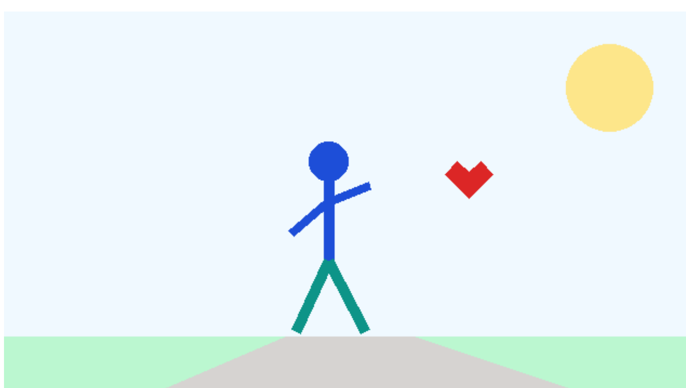

Figure 1. Conceptual illustration: aerobic walking and cardiometabolic health.

1. Introduction

Type 2 diabetes mellitus affects an estimated 1 in 10 adults worldwide and is a leading contributor to cardiovascular disease, renal failure, and premature mortality. Glycemic control, commonly assessed by glycated hemoglobin (HbA1c), is central to reducing long-term complications. Each 1-percentage-point reduction in HbA1c has been associated with substantial decreases in microvascular risk.

While pharmacotherapy is effective, lifestyle modification—particularly regular aerobic activity—offers complementary benefits with minimal cost and few adverse effects. Walking is accessible, requires no specialized equipment, and is well suited to middle-aged and older adults. However, unstructured advice to “exercise more” often yields poor adherence. We therefore designed a structured, progressively intensifying walking program and tested it against usual care.

1.1 Objectives

The primary objective was to determine the effect of a 12-week walking program on HbA1c. Secondary objectives were to assess effects on anthropometric and cardiometabolic measures and to evaluate program adherence and safety.

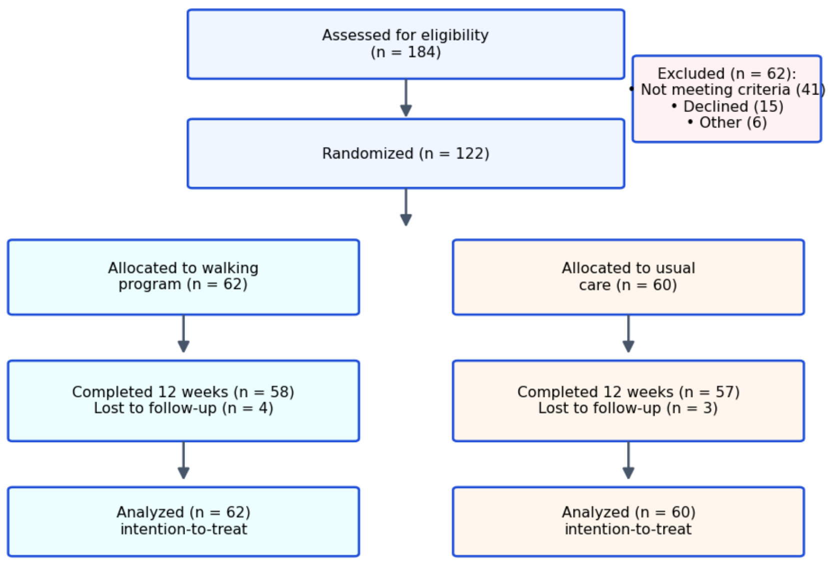

Figure 2. CONSORT participant flow diagram showing screening, randomization, follow-up, and analysis.

2. Methods

2.1 Design and Participants

This was a single-center, parallel-group, open-label randomized controlled trial conducted over six months. Eligible participants were adults aged 40–68 years with physician-diagnosed type 2 diabetes and a screening HbA1c of 7.5%–9.0%. Key exclusion criteria included unstable cardiovascular disease, mobility-limiting conditions, and recent (<3 months) changes to glucose-lowering medication.

2.2 Intervention

Participants in the intervention arm followed a supervised walking program targeting 150 minutes per week, divided across five sessions. Intensity progressed from moderate (40–50% heart-rate reserve) in weeks 1–4 to brisk (50–65%) by week 12. Sessions were logged with wearable step counters, and a coach provided weekly telephone support. The usual-care arm received standard lifestyle advice without structured supervision.

2.3 Outcomes and Statistics

The primary outcome was change in HbA1c from baseline to week 12. Secondary outcomes were measured at weeks 0, 6, and 12. Between-group differences were assessed using analysis of covariance adjusted for baseline values, with intention-to-treat as the primary analysis set. A two-sided p-value below 0.05 was considered significant.

Table 1. Baseline characteristics of participants

Characteristic	Walking (n=62)	Usual care (n=60)
Age, years (mean ± SD)	56.4 ± 7.1	55.9 ± 6.8
Female, n (%)	33 (53%)	31 (52%)
HbA1c, % (mean ± SD)	8.1 ± 0.4	8.0 ± 0.4
Body weight, kg	84.7 ± 12.3	83.9 ± 11.8
BMI, kg/m^2	30.2 ± 3.6	29.9 ± 3.4
Systolic BP, mmHg	134 ± 11	133 ± 12
Diabetes duration, years	6.2 ± 3.1	6.5 ± 3.4
On metformin, n (%)	55 (89%)	53 (88%)

SD, standard deviation; BMI, body mass index; BP, blood pressure. Groups were well balanced at baseline.

3. Results

Of 184 individuals screened, 122 were randomized and 115 (94%) completed the 12-week assessment. Baseline characteristics were balanced between arms (Table 1). Median adherence to prescribed walking sessions was 84%.

3.1 Primary Outcome

Mean HbA1c fell from 8.1% to 7.2% in the walking group and from 8.0% to 7.9% in usual care. The adjusted between-group difference at week 12 was –0.8 percentage points (95% CI –1.0 to –0.6, p<0.001), with separation evident by week 6.

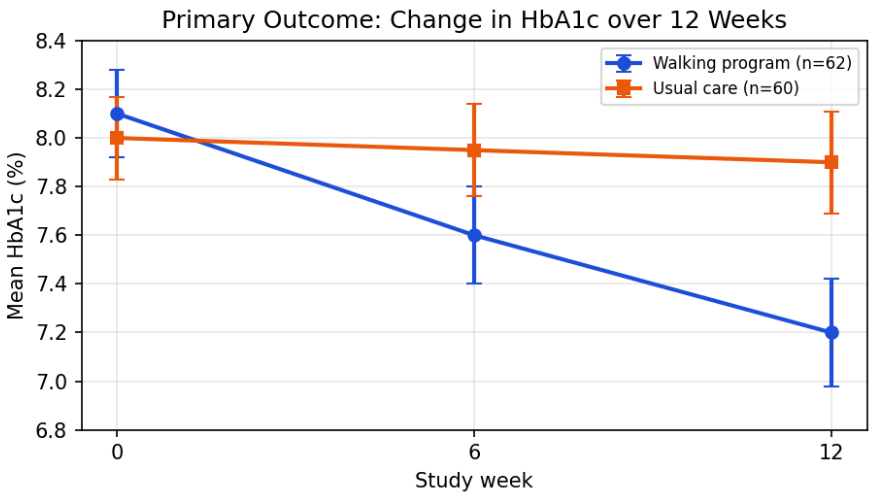

Figure 3. Mean HbA1c by study week; error bars denote standard error.

3.2 Secondary Outcomes

The walking group showed greater reductions across all secondary measures, including body weight, BMI, systolic blood pressure, and fasting glucose, as summarized below.

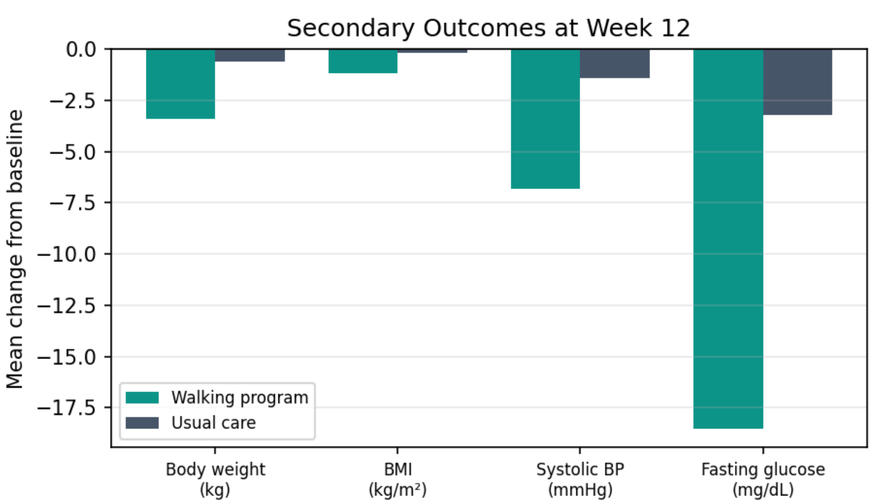

Figure 4. Mean change from baseline in secondary outcomes at week 12.

4. Discussion

In this trial, a structured 12-week walking program produced a clinically meaningful 0.8-percentage-point greater reduction in HbA1c than usual care. The magnitude of effect is comparable to that achieved by adding some oral glucose-lowering agents, but without pharmacologic cost or side-effect burden. Concurrent improvements in weight, blood pressure, and fasting glucose suggest broader cardiometabolic benefit.

Several features may explain the program's success. Progressive intensity allowed deconditioned participants to build capacity gradually, while weekly coaching and objective step tracking supported the high 84% adherence. These elements are low-cost and readily transferable to primary-care and community settings.

4.1 Limitations

This was a single-center, open-label study with a 12-week horizon; longer follow-up is needed to confirm durability of the glycemic benefit and to assess hard clinical endpoints. The synthetic nature of the data presented here also limits any generalization. Blinding of participants was not feasible given the behavioral intervention.

5. Conclusion

A simple, supervised walking program meaningfully improved glycemic control and cardiometabolic risk factors over 12 weeks. Given its low cost and high adherence, such programs warrant evaluation in larger, multi-center trials with longer follow-up.

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